Functional Dependency Discovery Pipeline – Report

1. Pipeline Design

The implemented pipeline is a fully automated functional dependency (FD) discovery and preparation workflow. It starts from raw datasets in CSV or .data format and applies two classical FD discovery algorithms: TANE and FASTFD. The workflow consists of three main stages: (1) FD discovery using Python implementations of TANE or FASTFD, (2) parsing and statistical analysis of discovered FDs, and (3) preparation of pipeline-ready outputs for downstream hybrid or LLM-based validation. All datasets are processed in batch mode, ensuring scalability and reproducibility.

2. Findings

- Dataset **nursery** using **TANE**: Success = True, Number of discovered FDs = 0.

Overall, the experiments showed that several datasets did not exhibit strong functional dependencies under the chosen constraints (maximum LHS size = 3). This is expected for datasets dominated by categorical or weakly dependent attributes. The pipeline successfully handled all datasets uniformly and produced consistent outputs for both algorithms.

3. Mistakes and Lessons Learned

During development, several issues were encountered. First, the initial design relied on external tools such as Metanome, which introduced installation and automation difficulties. This was resolved by implementing TANE and FASTFD directly in Python. Second, dataset format inconsistencies (.data vs .csv) caused silent failures in early stages; enforcing standardized input handling fixed this issue. Finally, the pipeline initially failed when no FDs were discovered because output files were not created. This was corrected by ensuring that output files are always generated, even when the FD set is empty. These mistakes led to a more robust and fully automated pipeline.